Distribution of Head Acceleration Events Varies by Position and Play Type in North American Football

Taylor A. Lee, BS,* Roy J. Lycke, PhD,† Patrick J. Lee,* Caroline M. Cudal,† Kelly J. Torolski,* Sean E. Bucherl, BS,† Nicolas Leiva-Molano, MS,† Paul S. Auerbach, MD, MS,‡§ Thomas M. Talavage, PhD,†¶ and Eric A. Nauman, PhD*†‖

Abstract

Objective: The goal of this pilot study was to evaluate the number of head acceleration events (HAEs) based on position, play type, and starting stance. **Design:** Prospective cohort study. **Setting:** Postcollegiate skill development camp during practice sessions and 1 exhibition game. **Participants:** Seventy-eight male adult North American football athletes. **Independent Variables:** A position was assigned to each participant, and plays in the exhibition game were separated by play type for analysis. During the exhibition game, video data were used to determine the effects of the starting position ("up" in a 2-point stance or "down" in a 3- or 4-point stance) on the HAEs experienced by players on the offensive line. **Main Outcome Measures:** Peak linear acceleration and number of HAEs greater than 20 g (g = 9.81 m/s^2) were measured using an xPatch (X2 Biosystems, Seattle, WA). **Results:** Four hundred thirty-seven HAEs were recorded during practices and 272 recorded during the exhibition game; 98 and 52 HAEs, the greatest number of HAEs by position in the game, were experienced by the offensive and defensive linemen, respectively. Linebackers and tight ends experienced high percentages of HAEs above 60 g. Offensive line players in a down stance had a higher likelihood of sustaining a HAE than players in an up stance regardless of the type of play (run vs pass). **Conclusions:** Changing the stance of players on the offensive line and reducing the number of full-contact practices will lower HAEs.

Key Words: traumatic brain injury, offensive linemen, run versus pass, player stance

(*Clin J Sport Med* 2020;00:1–6)

INTRODUCTION

Traumatic brain injuries (TBIs) result from both direct head impacts and whiplash motions[1] that can result in diminished cognitive, motor, and sensory functions, and in severe cases, permanent brain damage.[2] In 2010, it was estimated the lifetime costs (direct medical and indirect) of all TBI in the United States was approximately $76.5 billion.[3] There is growing concern about the long-term effects playing North American football can have on a player's brain, particularly long-term cognitive health. This problem was first identified and documented in football players when former National Football League players' autopsies revealed they had a neurodegenerative disease known as chronic traumatic encephalopathy (CTE), previously observed only in retired pugilists.[4–8] Patients with CTE can exhibit short-term memory deficiency, lack of emotional stability, poor decision making, difficulty with behavioral control, and suicidal tendencies.[9] Chronic traumatic encephalopathy has been confirmed postmortem in more than 120 former professional, 48 college, and 4 high school American football players, although selection bias in the studies may contribute to the high observation rate of those confirmed to have CTE.[4,5,8,10–14] Development of CTE does not correlate with history of diagnosed concussions,[15] a type of TBI, but rather with a history of repetitive brain trauma.[16–21] This suggests that continuous exposure to subconcussive impacts, or impacts that do not cause a diagnosed concussion, may be the critical aspect contributing to CTE.[9,16–18,22,23]

This causation is consistent with the results of studies by Talavage et al, who determined that over the course of a season, high school football players experience impaired neurocognitive and neurophysiological capabilities without being diagnosed with concussion or displaying any classic concussion symptoms.[24,25] Specifically, these investigators found that subconcussive blows can have effects over an extended period of time similar to those from a concussion.[19,24–34] It has been established that the number and locations of impacts are important factors to consider when determining whether a player is going to experience decreased neurocognitive function by the end of a season.[24,25]

Head acceleration events (HAEs) are direct blows to the head and/or whiplash motions of the head associated with blows to other parts of the body. Studies examining changes in brain function, chemistry, and structure after periods of exposure to repeated HAEs reveal that although high-magnitude individual HAEs are likely to produce near-term clinical symptoms and might contribute to chronic injury, these "big hits" are not the sole major cause of later-life neurodegenerative disorders in former collision-sport athletes.[16,35]

Submitted for publication January 29, 2019; accepted June 19, 2019.

*From the *Department of Mechanical Engineering, Purdue University, West Lafayette, Indiana; †Department of Biomedical Engineering, Weldon School of Biomedical Engineering, Purdue University, West Lafayette, Indiana; ‡Military/Emergency Medicine, Uniformed Services University of the Health Sciences, Bethesda, Maryland; §Department of Emergency Medicine, Stanford University School of Medicine, Stanford, California; ¶Department of Electrical and Computer Engineering, Purdue University, West Lafayette, Indiana; and ‖Department of Basic Medical Sciences, Purdue University, West Lafayette, Indiana.*

T. A. Lee is currently receiving a grant (DGE-1333468) from National Science Foundation Graduate Research Fellowships Program. The remaining authors report no conflicts of interest.

Corresponding Author: Eric A. Nauman, PhD, School of Mechanical Engineering, 3061A Mechanical Engineering Building, 585 Purdue Mall, West Lafayette, IN 47907 (enauman@purdue.edu).

Copyright © 2020 Wolters Kluwer Health, Inc. All rights reserved.

http://dx.doi.org/10.1097/JSM.0000000000000778

Copyright © 2020 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

Numerous studies have reported abnormal neuroimaging and neurocognitive measurements in athletes who experienced appreciable numbers of HAEs.[26,36–40] Examples include within-season functional magnetic resonance imaging (fMRI) changes that are strongly associated with the number of HAEs in the preceding week[24] and post-season cognitive testing and fMRI changes predicted by average weekly number of HAEs during the season.[41] These investigations suggest that cumulative mechanical loading might result in more injury to brain tissues than can be repaired by natural processes,[35] with the limit likely lying between 65 and 90 HAEs per week for high school football players.[41] The concept of a "cumulative subconcussive threshold" has been extrapolated by Alosco et al to predict how lifetime history of exposure to HAEs might predict later-life risk of cognitive dysfunction and structural damage.[42,43] Reducing the total number of HAEs experienced can be achieved through 4 strategies: (1) modify the rules of the game to limit head impacts,[17,44] (2) teach and use better techniques during practice, (3) reduce or eliminate full-contact practice activities,[17] which might translate to the style of play in games, and (4) develop helmets that absorb more energy.[17] Of these 4, rules modification and eliminating full-contact practices can be implemented immediately. One specific rule change that has been suggested is to eliminate the 3- and 4-point ("down") stances for linemen, as these stances may increase the number of collision and other head impacts sustained due to the initial weight distribution of the player and height of the head.[17] Consequently, the goal of this pilot study was to evaluate the potential for this rule change to reduce HAEs in a postcollegiate group of athletes participating in a football skill development and professional recruitment camp.

METHODS

Subjects

The study was approved by the Purdue Institutional Review Board. Written informed consent was obtained from each subject before enrolling in the study. Football (FB) athletes participating in up to 3 practice sessions and 1 exhibition game as part of a postcollegiate skill development camp were recruited for the study. The position of each participant was recorded and divided into the following categories: defensive backs (DBs), defensive linemen (DL), linebackers (LB), offensive linemen (OL), tight ends (TEs), running backs (RB), wide receivers (WRs), and skill players (quarterbacks [QB], placekickers [K], and punters [P]).

Impact Data Collection and Analysis

HAEs were monitored using the xPatch (X2 Biosystems, Seattle, WA). Each practice and each game were considered a separate "session." Sensors were placed on each participant for all practice sessions and for the exhibition game. A sensor was affixed behind the player's right ear with an adhesive patch after cleaning the area with isopropyl alcohol.[45,46] Each head impact was recorded as a separate "event" on the sensor when the peak linear acceleration (PLA) projected on any axis was greater than 10 g (g = 9.81 m/s^2). Data were downloaded using the Head Impact Monitoring System software (X2 Biosystems) after each session. For the purposes of this study, the clack recognition algorithm (X2 Biosystems) was not used because events caused by different mechanisms (eg, direct impacts, whiplashes, falls, and dives) were not distinguished.[45,46]

Data from the sensors were processed using a custom MATLAB (MathWorks, Natick, MA) program to include only events occurring within the valid time window of each session and registering a PLA greater than or equal to 20 g. Although the sensors were programmed to collect low-acceleration events (10-20 g), these events were excluded from analysis because they can frequently result from nonimpact mechanisms (eg, changing direction, kicking the ball, and stopping).[46] Data were also analyzed to remove sessions that exhibited evidence of a defective sensor or detached sensor.

Video Data Collection and Analysis

Video footage of the exhibition game was collected using 6 Sony HDR-CX160 AVCHD HD Handycam Camcorders and 2 Sony DCR-SX85 Handycam Camcorders (Sony Corporation, Kōnan, Minato, Tokyo). The HD cameras were set to record on long time with HD quality. Three HD cameras were placed on each side of the field. During the game play, 1 camera covered the middle of the field from one 30 yardline to the other 30 yardline, whereas the other 2 cameras per sideline each covered the area from one 30 yardline to the closest end zone. The 2 DCR-SX85 cameras were operated at the field level.

For each play examined, there were 5 OL players positioned at the line of scrimmage. Depending on the play, the TE would either be positioned at the line of scrimmage directly next to the OL or he would be positioned further away in the slot receiver position. If the TE was positioned directly next to an OL, then the TE was considered a player on the offensive line and eligible for stance analysis.

The video footage revealed which players were on the field for each play and the stance of players positioned as the offensive line (up [2-point] or down [3- or 4-point], Figure 1). The impact data for the game were filtered to only include impacts for players who were on the field for each play. The hit rate for the players positioned as the offensive line was calculated for the ith play type (run or pass) and jth stance (up or down) by,

$$\text{player hit rate}_{ij} = \frac{\sum \text{players registering HAE}_{ij}}{\sum \text{players}_{ij}}.$$

Dividing the total number of players who sustained hits for a specific combination of the play type and stance by the total number of players with that specific combination makes it possible for rates to be compared by normalizing by the respective sample sizes.

Statistics

Statistics were performed using Stata/SE 15 (StataCorp LLC, College Station, TX). Histograms were generated by binning the HAEs by PLA in 20-g intervals and were separated by position. They were compared using a χ^2 test, with post hoc testing performed using pairwise χ^2 tests with a Bonferroni correction. The number of hits was compared using the Wilcoxon rank-sum test. Hit rates for the players on the offensive line were compared using a χ^2 test, with post hoc

Copyright © 2020 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

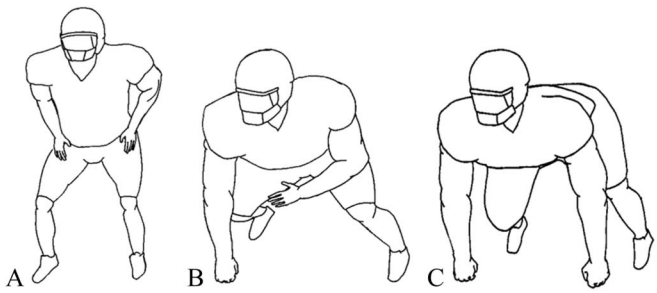

Figure 1. The players on the offensive line began each play in a (A) 2-point, (B) 3-point, or (C) 4-point stance. A 2-point stance was considered the "up" stance and a 3-point or 4-point stance was considered a "down" stance.

testing performed using pairwise χ^2 tests with a Bonferroni correction.

RESULTS

Among the 78 participants, there was a total of 141 measured player practice sessions and 66 player game sessions (defective sensors: n = 2; detached: n = 35). A total of 437 (62%) HAEs were recorded during practices, and 272 (38%) were recorded during the exhibition game (Table), 265 of which occurred during a run or pass play.

The histogram of HAEs for practice sessions (Figure 2) and the exhibition game (Figure 3) demonstrated significant differences between the different positions for both the practices ($P = 0.011$) and game ($P = 0.017$). For practices, there were significant differences between RB and OL ($P < 0.001$), and RB and DL ($P < 0.001$). For the game, there were significant differences between OL and LB ($P = 0.001$), and OL and TE ($P < 0.001$).

The number of HAEs that exceeded 20 g observed over the ensemble of players on the field differed appreciably between passing and running plays (Figure 4). There were 46 pass plays and 23 run plays in the exhibition game. The pass plays had a total of 242 players on the offensive line (90 up and 152 down); the run plays had a total of 119 players on the offensive line (28 up and 91 down). The frequency of players in the up stance was significantly greater for pass plays than for run plays (χ^2 test, $P = 0.009$). When a play was scripted to be a run, the number of HAEs observed to exceed 20 g (5.57 ± 3.00 HAEs per play) was nearly twice that observed when a play was scripted to be a pass (2.98 ± 1.65 HAEs per play). The Wilcoxon rank-sum test revealed these distributions to be significantly different ($P < 0.001$).

Regardless of play type, players in a down stance had a higher likelihood of sustaining a HAE than players starting in an up stance (Figure 5; χ^2 test, $P = 0.001$). When separating by play type, players in a down stance for a run play sustained significantly more HAEs than players in an up stance (χ^2 test, $P < 0.001$) or down stance (χ^2 test, $P = 0.003$) for a pass.

DISCUSSION

The purpose of this pilot study was to examine the HAE characteristics for players participating in a postcollegiate skill and recruitment football showcase. Analysis by position revealed that TEs are of particular concern because they can act as both blockers and receivers for the same play, increasing their odds for sustaining a HAE. The starting stance of the offensive line players influences the frequency at which these players accumulate HAEs, with players starting in the down stance recording a higher hit rate than those starting in the up stance. The findings from this study are intended to aid in developing potential techniques and rule changes to decrease the accumulation of HAEs in football athletes, notably linemen.

The hit distributions in practices and in games varied by position. The LBs and TEs tended to sustain the greatest number of high-magnitude impacts (above 60 g PLA; 9.68 and 18.75%, respectively, in games). Unlike linemen, linebackers have a defensive role in the game and position on the field that encourage them to collide with offensive players head on or at an oblique angle and at a high speed of approach. Tight ends had the greatest frequency of high-magnitude game hits compared with other player positions. Tight ends often function as both an OL (to block) and as a WR (to catch a pass) within the same play. Blocking generates a large number of lower magnitude HAEs,[47,48] and receiving passes followed by being tackled generates high-magnitude impacts.

Offensive and defensive linemen deliver and receive the most hits during sessions, but they tend to be smaller in magnitude (20-40 g PLA). Players in these positions typically lack a running start before contact, leading to lower PLA

Table. Head acceleration events by Position as a Function of Practices (Session 1, 2, and 3) and Game

Position	No. of Players	Practice HAEs	Game HAEs	Total HAEs
Defensive backs	19	22 (0, 8, 14)	36	58
Defensive linemen	9	96 (0, 50, 46)	52	148
Linebackers	10	45 (0, 14, 31)	31	76
Offensive linemen	10	155 (0, 76, 79)	98	253
Tight ends	3	41 (1, 34, 6)	16	57
Running backs	8	41 (0, 13, 28)	21	62
Wide receivers	12	28 (3, 15, 10)	11	39
Skill	7	9 (0, 7, 2)	7	16
Quarterbacks	4	6 (0, 5, 1)	7	13
Placekickers/Punters	3	3 (0, 2, 1)	0	3

Copyright © 2020 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

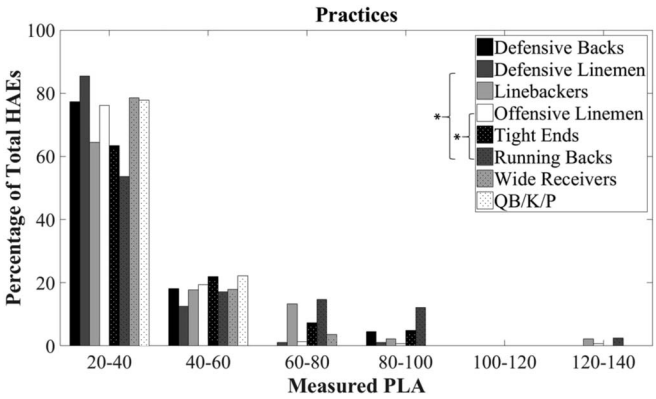

Figure 2. Head acceleration event distribution by position (DB, DL, LB, OL, TE, RB, WR, and QB/K/P) for all 3 practice sessions.

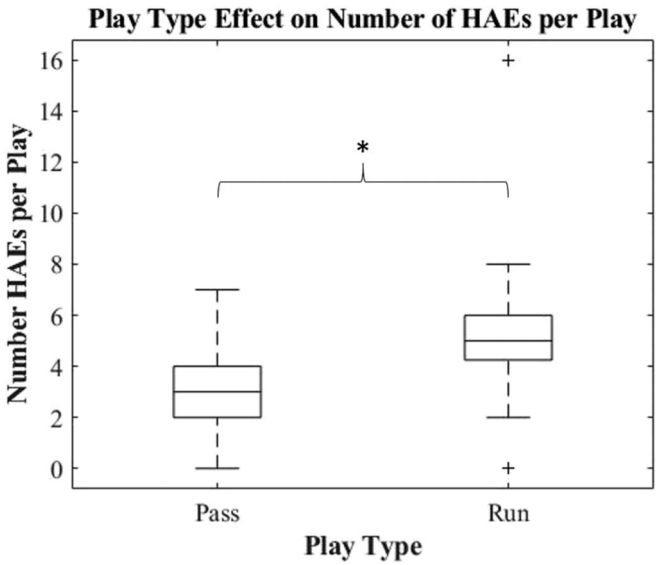

Figure 4. Comparison of the average number of play-related HAEs (exceeding 20 g) observed across all monitored players on the field during plays intended to be passing versus running.

measures. This is similar to trends seen in other studies.[47,49] Although the impacts are smaller in magnitude, these players may still be at risk of developing neurological changes because the number of HAEs may be a factor in the severity and degree of neurological changes experienced.[4,5,19,24–26,37,41,50] It is possible that these less forceful HAEs are the root cause of CTE.[22,23]

The average number of HAEs (exceeding 20 g) per play was significantly lower for pass plays than run plays. This is expected for linemen because the OL are more likely to generate impacts when they act as aggressors for run plays than when they play the protector role for pass plays. This trend was generally true for players at all positions, not just linemen. So, although linemen dominate the number of impacts for both pass and run plays, pass plays in general result in less contact between opposing players at all positions.

The initial stance of the OL and TE for pass and run plays affects the hit rate. Players who began in the down stance generally were more likely to sustain HAEs, regardless of the play type. Most line players end up blocking from an up stance, so beginning play in a 2-point stance has the added benefit of making the head less vulnerable to a substantial HAE. There was a greater frequency of players in the up stance for pass plays than for run plays. Players who started in a down stance for a run play had a significantly higher hit rate than players in either the up or down stance for pass plays. During run plays, OL are tasked not only with protecting the quarterback but also with moving DL to clear a path for the RB. This requires the OL to keep their center of gravity low, which puts the head in a position that can contribute to a higher hit rate.

Limitations of this article include some nominal variability in the measurements obtained from the xPatch sensor.[45] The RMS and mean absolute errors for the xPatch are both typically 20%, which is sufficient for

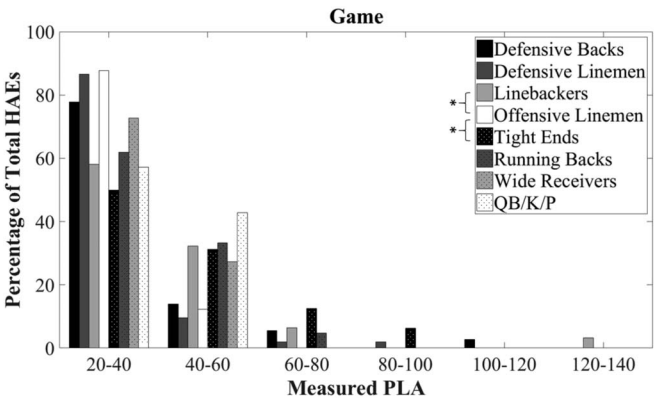

Figure 3. Head acceleration event distribution by position (DB, DL, LB, OL, TE, RB, WR, and QB/K/P) for the exhibition game.

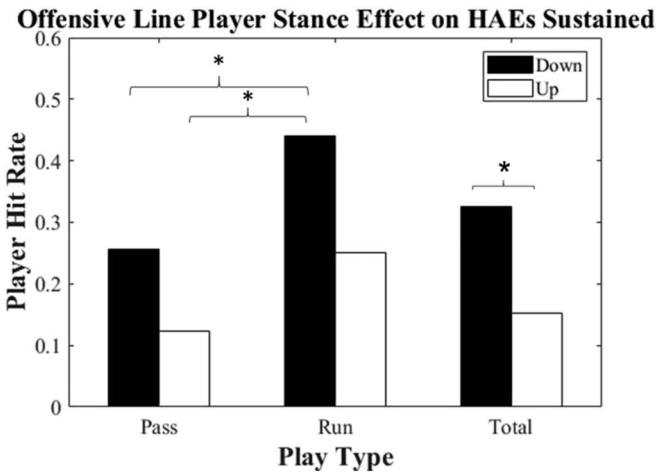

Figure 5. Comparison of the rate that OL and TEs sustain impacts separated by play type (pass vs run) and stance (up vs down).

Copyright © 2020 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

characterizing differences in HAE distributions and cumulative exposure.[45] In addition, this study only includes 2 full practice sessions and 1 game, resulting in a relatively small sample size. However, the results and trends documented here are sufficient to warrant a larger study to determine whether the trends hold. The small sample size of players in the TE position poses an issue for the validity of the trends recorded in this study. These trends warrant further study to determine whether and how HAEs can be reduced for players in this position.

CONCLUSIONS

The starting stance of the line players influences the frequency at which these players accumulate HAEs. Players starting in the down stance have a higher HAE rate than players starting in the up stance. Changing linemen stance rules and reducing the number of full-contact practices will lower the number of HAEs sustained and increase player safety. Tight ends are of particular concern because they can act as both blockers and receivers for the same play, increasing the chance of sustaining impact. We plan a study of longer duration to further investigate these contentions.

ACKNOWLEDGMENTS

The authors thank Nathaniel and Anneliese Nauman for their assistance in this study.

References

1. Menon DK, Schwab K, Wright DW, et al. Position statement: definition of traumatic brain injury. *Arch Phys Med Rehabil*. 2010;91:1637–1640.
2. DeKosky ST, Ikonomovic MD, Gandy S. Traumatic brain injury—football, warfare, and long-term effects. *N Engl J Med*. 2010; 363:1293–1296.
3. Coronado V, McGuire L, Faul M, et al. Traumatic brain injury epidemiology and public health issues. In: Zasler N, Katz DI, Zafonte R, eds *Brain Injury Medicine: Principles and Practices*. New York, NY: Demos Medical Publishing, LLC; 2012.
4. Omalu BI, DeKosky ST, Hamilton RL, et al. Chronic traumatic encephalopathy in a national football league player: part II. *Neurosurgery*. 2006;59:1086–1092; discussion 1092-1083.
5. Omalu BI, DeKosky ST, Minster RL, et al. Chronic traumatic encephalopathy in a national football league player. *Neurosurgery*. 2005;57:128–134; discussion 128-134.
6. Martland H. Punch drunk. *JAMA*. 1928;91:1103–1107.
7. Corsellis JA, Bruton CJ, Freeman-Browne D. The aftermath of boxing. *Psychol Med*. 1973;3:270–303.
8. Omalu BI, Hamilton RL, Kamboh MI, et al. Chronic traumatic encephalopathy (CTE) in a National Football League Player: case report and emerging medicolegal practice questions. *J Forensic Nurs*. 2010;6:40–46.
9. Stern RA, Riley DO, Daneshvar DH, et al. Long-term consequences of repetitive brain trauma: chronic traumatic encephalopathy. *PM R*. 2011; 3:S460–S467.
10. Mez J, Daneshvar DH, Kiernan PT, et al. Clinicopathological evaluation of chronic traumatic encephalopathy in players of American football. *JAMA*. 2017;318:360–370.
11. Omalu B, Bailes J, Hamilton RL, et al. Emerging histomorphologic phenotypes of chronic traumatic encephalopathy in American athletes. *Neurosurgery*. 2011;69:173–183.
12. Omalu B, Small GW, Bailes J, et al. Postmortem autopsy-confirmation of antemortem [F-18] FDDNP-PET scans in a football player with chronic traumatic encephalopathy. *Neurosurgery*. 2017;82:237–246.
13. Omalu BI, Bailes J, Hammers JL, et al. Chronic traumatic encephalopathy, suicides and parasuicides in professional American athletes: the role of the forensic pathologist. *Am J Forensic Med Pathol*. 2010;31:130–132.
14. Maroon JC, Winkelman R, Bost J, et al. Chronic traumatic encephalopathy in contact sports: a systematic review of all reported pathological cases. *PLoS One*. 2015;10:e0117338.
15. Hazrati LN, Tartaglia MC, Diamandis P, et al. Absence of chronic traumatic encephalopathy in retired football players with multiple concussions and neurological symptomatology. *Front Hum Neurosci*. 2013;7:222.
16. McKee AC, Cantu RC, Nowinski CJ, et al. Chronic traumatic encephalopathy in athletes: progressive tauopathy after repetitive head injury. *J Neuropathol Exp Neurol*. 2009;68:709–735.
17. Gavett BE, Stern RA, McKee AC. Chronic traumatic encephalopathy: a potential late effect of sport-related concussive and subconcussive head trauma. *Clin Sports Med*. 2011;30:179–188, xi.
18. McKee AC, Stern RA, Nowinski CJ, et al. The spectrum of disease in chronic traumatic encephalopathy. *Brain*. 2013;136:43–64.
19. Bailes JE, Petraglia AL, Omalu BI, et al. Role of subconcussion in repetitive mild traumatic brain injury. *J Neurosurg*. 2013;119:1235–1245.
20. Bailes JE, Turner RC, Lucke-Wold BP, et al. Chronic traumatic encephalopathy: is it real? The relationship between neurotrauma and neurodegeneration. *Neurosurgery*. 2015;62:15–24.
21. Turner RC, Lucke-Wold BP, Robson MJ, et al. Alzheimer's disease and chronic traumatic encephalopathy: distinct but possibly overlapping disease entities. *Brain Inj*. 2016;30:1279–1292.
22. Baugh CM, Stamm JM, Riley DO, et al. Chronic traumatic encephalopathy: neurodegeneration following repetitive concussive and subconcussive brain trauma. *Brain Imaging Behav*. 2012;6:244–254.
23. Tagge CA, Fisher AM, Minaeva OV, et al. Concussion, microvascular injury, and early tauopathy in young athletes after impact head injury and an impact concussion mouse model. *Brain*. 2018;141:422–458.
24. Talavage TM, Nauman EA, Breedlove EL, et al. Functionally-detected cognitive impairment in high school football players without clinically-diagnosed concussion. *J Neurotrauma*. 2014;31:327–338.
25. Breedlove EL, Robinson M, Talavage TM, et al. Biomechanical correlates of symptomatic and asymptomatic neurophysiological impairment in high school football. *J Biomech*. 2012;45:1265–1272.
26. Abbas K, Shenk TE, Poole VN, et al. Alteration of default mode network in high school football athletes due to repetitive subconcussive mild traumatic brain injury: a resting-state functional magnetic resonance imaging study. *Brain Connect*. 2015;5:91–101.
27. Abbas K, Shenk TE, Poole VN, et al. Effects of repetitive sub-concussive brain injury on the functional connectivity of Default Mode Network in high school football athletes. *Dev Neuropsychol*. 2015;40:51–56.
28. Robinson ME, Shenk TE, Breedlove EL, et al. The role of location of subconcussive head impacts in FMRI brain activation change. *Dev Neuropsychol*. 2015;40:74–79.
29. Shenk TE, Robinson ME, Svaldi DO, et al. FMRI of visual working memory in high school football players. *Dev Neuropsychol*. 2015;40:63–68.
30. Johnson B, Neuberger T, Gay M, et al. Effects of subconcussive head trauma on the default mode network of the brain. *J Neurotrauma*. 2014; 31:1907–1913.
31. Johnson B, Zhang K, Gay M, et al. Alteration of brain default network in subacute phase of injury in concussed individuals: resting-state fMRI study. *Neuroimage*. 2012;59:511–518.
32. Stevens MC, Lovejoy D, Kim J, et al. Multiple resting state network functional connectivity abnormalities in mild traumatic brain injury. *Brain Imaging Behav*. 2012;6:293–318.
33. Zhou Y, Milham MP, Lui YW, et al. Default-mode network disruption in mild traumatic brain injury. *Radiology*. 2012;265:882–892.
34. Zhu DC, Covassin T, Nogle S, et al. A potential biomarker in sports-related concussion: brain functional connectivity alteration of the default-mode network measured with longitudinal resting-state fMRI over thirty days. *J Neurotrauma*. 2015;32:327–341.
35. Talavage TM, Nauman EA, Leverenz LJ. The role of medical imaging in the recharacterization of mild traumatic brain injury using youth sports as a laboratory. *Front Neurol*. 2015;6:273.
36. Bari S, Svaldi DO, Jang I, et al. Dependence on subconcussive impacts of brain metabolism in collision sport athletes: an MR spectroscopic study. *Brain Imaging Behav*. 2018;13:735–749.
37. Chun IY, Mao X, Breedlove EL, et al. DTI detection of longitudinal WM abnormalities due to accumulated head impacts. *Dev Neuropsychol*. 2015;40:92–97.
38. Poole VN, Abbas K, Shenk TE, et al. MR spectroscopic evidence of brain injury in the non-diagnosed collision sport athlete. *Dev Neuropsychol*. 2014;39:459–473.

Copyright © 2020 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

39. Svaldi DO, Joshi C, Robinson ME, et al. Cerebrovascular reactivity alterations in asymptomatic high school football players. *Dev Neuropsychol*. 2015;40:80–84.
40. Svaldi DO, McCuen EC, Joshi C, et al. Cerebrovascular reactivity changes in asymptomatic female athletes attributable to high school soccer participation. *Brain Imaging Behav*. 2017;11:98–112.
41. Nauman EA, Breedlove KM, Breedlove EL, et al. Post-season neurophysiological deficits assessed by ImPACT and fMRI in athletes competing in American football. *Dev Neuropsychol*. 2015;40:85–91.
42. Alosco ML, Kasimis AB, Stamm JM, et al. Age of first exposure to American football and long-term neuropsychiatric and cognitive outcomes. *Transl Psychiatry*. 2017;7:e1236.
43. Alosco ML, Tripodis Y, Jarnagin J, et al. Repetitive head impact exposure and later-life plasma total tau in former National Football League players. *Alzheimers Dement (Amst)*. 2017;7:33–40.
44. Auerbach PS, Waggoner WH II. It's time to change the rules. *JAMA*. 2016;316:1260–1261.
45. Cummiskey B, Schiffmiller D, Talavage TM, et al. Reliability and accuracy of helmet-mounted and head-mounted devices used to measure head accelerations. *J Sports Eng Technol*. 2017;231:144–153.
46. McCuen E, Svaldi D, Breedlove K, et al. Collegiate women's soccer players suffer greater cumulative head impacts than their high school counterparts. *J Biomech*. 2015;48:3720–3723.
47. Broglio SP, Eckner JT, Martini D, et al. Cumulative head impact burden in high school football. *J Neurotrauma*. 2011;28:2069–2078.
48. Broglio SP, Sosnoff JJ, Shin S, et al. Head impacts during high school football: a biomechanical assessment. *J Athl Train*. 2009;44:342–349.
49. Schnebel B, Gwin JT, Anderson S, et al. In vivo study of head impacts in football: a comparison of National Collegiate Athletic Association Division I versus high school impacts. *Neurosurgery*. 2007;60:490–496.
50. Breedlove KM, Breedlove EL, Robinson M, et al. Detecting neurocognitive and neurophysiological changes as a result of subconcussive blows in high school football athletes. *Athl Train Sports Health Care*. 2014;6:119–127.

Copyright © 2020 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.